

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 14A.3

IONIC PRODUCT OF WATER, KW

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	14: Acid-Base Equilibria
SUB-TOPIC:	14A.3: Ionic Product of Water, Kw
TOTAL QUESTIONS:	1 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	1 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 14A.3

Q1. Write the expression for the ionic product of water, K_w .

[1 Marks]

WCH14/01/Jan22/Q12(a)

SECTION B: WORKED MARK SCHEME — 14A.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Write the expression for the ionic product of water, K_w . $K_w = [H^+][OH^-]$	[1]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 14A.3**FAQ #1: How does temperature affect K_w ?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How does temperature affect K_w ?

MODEL ANSWER / EXAMINER EXPECTATION:

The dissociation of water is endothermic, so increasing temperature shifts equilibrium to the right, increasing K_w and decreasing the pH of pure water.